IE540
L10. Policy based RL : TRPO, PPO

KAIST ISE
Hayong Shin

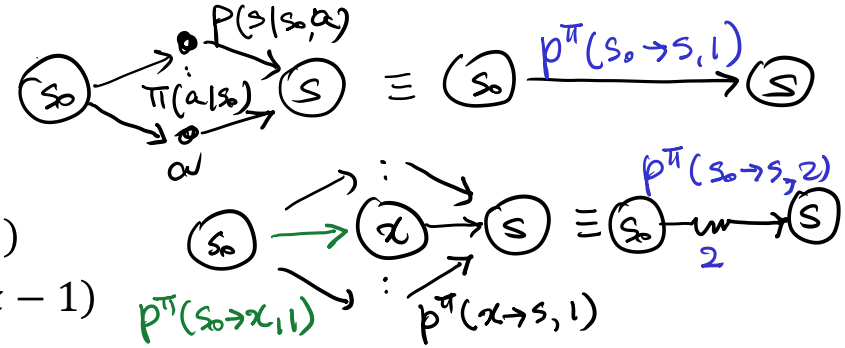
Policy based RL (Policy Optimization)

- **Policy Gradient**
  - Stochastic policy gradient
    - REINFORCE
    - Actor-Critic : Stochastic policy (actor) & Value function (critic)
      - Q A-C : Critic is Q(s,a)
      - Advantage A-C (A2C), Async. Adv. A-C (A3C, Deepmind, ICML2016)
      - Other variants : TD A-C, TD(λ) A-C, …
    - TRPO(ICML 2015), PPO(arXiv 2017), PPG(arXiv, 2020) : Schulman
    - SAC (Haarnoja, ICML 2018)
  - Deterministic policy gradient
    - DPG(Silver, ICML 2014)
    - DDPG(Deepmind, ICLR 2016)
    - TD3 (Fujimoto, ICML 2018)
- **Derivative Free Optimization**
  - Evolutionary strategy (ES) : population based
    - CEM (Cross-Entropy Method)
    - GA (Genetic Algorithm), SA (Simulated Annealing)
- **PG + ES** : Combine PG with ES
  - ERL (NIPS 2018) : GA + DDPG
  - CEM-RL (ICLR 2019) : CEM + TD3
  - …

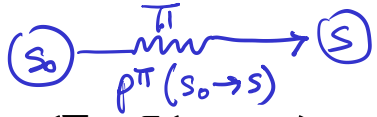
Discounted stationary distribution

$p^\pi(s_0 \to s, k+\ell) = \sum_x p^\pi(s_0 \to x, k)\, p^\pi(x \to s, \ell)$ Chapman-Kolmogorov

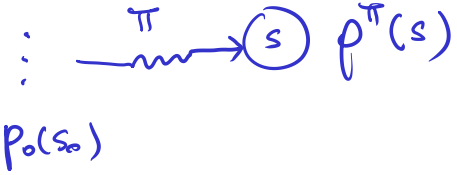
- Let $p^\pi(s_0 \to s, k)$: prob. of state after k steps from s_0 under π
  - $p^\pi(s_0 \to s, 0) = \begin{cases} 1, \text{if } s = s_0 \\ 0, \text{otherwise} \end{cases}$
  - $p^\pi(s_0 \to s, 1) = \sum_a \pi(a|s_0) p(s|s_0, a)$
  - $p^\pi(s_0 \to s, 2) = \sum_x p^\pi(s_0 \to x, 1)\, p^\pi(x \to s, 1)$
  - $p^\pi(s_0 \to s, k) = \sum_x p^\pi(s_0 \to x, 1)\, p^\pi(x \to s, k-1)$



- $\rho^\pi(s_0 \to s) = \sum_{k=0}^{\infty} \gamma^k\, p^\pi(s_0 \to s, k)$
  - improper discounted stationary distribution from s_0 $(\sum_s \rho^\pi(s_0 \to s) \neq 1)$
  - a.k.a. discounted visitation frequency



- $\rho^\pi(s) = \sum_{s_0} p_0(s_0) \sum_{k=0}^{\infty} \gamma^k\, p^\pi(s_0 \to s, k)$
  - $p_0(s_0)$: initial state distribution
  - improper discounted stationary distribution $(\sum_s \rho^\pi(s) \neq 1)$



- $d^\pi(s) = \dfrac{\rho^\pi(s)}{\sum_{x \in \mathcal{S}} \rho^\pi(x)}$
  - (proper) discounted stationary distribution $(\sum_s d^\pi(s) = 1)$

Value functions

- State value function, action value function
  - $Q^{\pi}(s_t, a_t) = \mathbb{E}_{\tau_{t+1}\sim\pi}\left[\sum_{k=0}^{\infty}\gamma^k r_{t+k} \mid s_t, a_t\right]$
  - $V^{\pi}(s_t) = \mathbb{E}_{a_t\sim\pi(.|s_t)}\left[Q^{\pi}(s_t, a_t)\right] = \mathbb{E}_{a_t\sim\pi(.|s_t),\tau_{t+1}\sim\pi}\left[\sum_{k=0}^{\infty}\gamma^k r_{t+k} \mid s_t\right]$
- Bellman expectation eq.
  - $V^{\pi}(s_t) = \mathbb{E}_{a_t\sim\pi(.|s_t),r_t,s_{t+1}\sim p(.|s_t,a_t)}\left[r_t + \gamma V^{\pi}(s_{t+1})\right]$
  - $Q^{\pi}(s_t, a_t) = \mathbb{E}_{r_t,s_{t+1}\sim p(.|s_t,a_t)}[r_t + \gamma V^{\pi}(s_{t+1})]$
    $= \mathbb{E}_{r_t,s_{t+1}\sim p(.|s_t,a_t),\mathbf{a}_{t+1}\sim\pi(.|s_{t+1})}[r_t + \gamma Q^{\pi}(s_{t+1}, \mathrm{a}_{t+1})]$
- Advantage function
  - $A^{\pi}(s_t, a_t) = Q^{\pi}(s_t, a_t) - V^{\pi}(s_t)$
    $= \mathbb{E}_{r_t,s_{t+1}\sim p(.|s_t,a_t)}[r_t + \gamma V^{\pi}(s_{t+1}) - V^{\pi}(s_t)]$ (TD advantage)
  - TD advantage estimator
    - $A_t = \delta_t^V = r_t + \gamma V_v(s_{t+1}) - V_v(s_t)$
  - GAE (Generalized Adv. Estimator) $\approx$ TD(λ) advantage estimator
    - $A_t = \sum_{l=0}^{T-t-1}(\gamma\lambda)^l\,\delta_{t+l}^V$ (see next slide)

GAE

- GAE(Generalized Adv. Estimator) by Schulman et al (ICLR, 2016)
- Definition
  - 1-step TD residual w.r.t. V: $\delta_t^V = r_t + \gamma V(s_{t+1}) - V(s_t)$
  - k-step TD advantage $\hat{A}_t^{(k)} \equiv \sum_{l=0}^{k-1} \gamma^l \, \delta_{t+l}^V$
    - $\hat{A}_t^{(1)} = \delta_t^V = -V(s_t) + r_t + \gamma V(s_{t+1})$
    - $\hat{A}_t^{(2)} = \delta_t^V + \gamma\delta_{t+1}^V = -V(s_t) + r_t + \gamma r_{t+1} + \gamma^2 V(s_{t+2})$
    - $\hat{A}_t^{(3)} = \delta_t^V + \gamma\delta_{t+1}^V + \gamma^2\delta_{t+2}^V = -V(s_t) + r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \gamma^3 V(s_{t+3})$
    - $\cdots$
    - $\hat{A}_t^{(k)} = \sum_{l=0}^{k-1} \gamma^l \, \delta_{t+l}^V = -V(s_t) + r_t + \gamma r_{t+1} + \cdots + \gamma^{k-1} r_{t+k-1} + \gamma^k V(s_{t+k})$
    - $\hat{A}_t^{(\infty)} = -V(s_t) + \sum_{l=0}^{\infty} \gamma^l \, r_{t+l}$: empirical return $-V(s_t)$
  - GAE (Generalized Advantage Estimator)
    - $\hat{A}_t^{\mathrm{GAE}} \equiv (1-\lambda)\left(\hat{A}_t^{(1)} + \lambda\hat{A}_t^{(2)} + \lambda^2\hat{A}_t^{(3)} + \cdots\right)$: weighted average of $\hat{A}_t^{(k)}$

$$
\begin{aligned}
&= (1-\lambda)(\delta_t^V + \lambda(\delta_t^V + \gamma\delta_{t+1}^V) + \lambda^2(\delta_t^V + \gamma\delta_{t+1}^V + \gamma^2\delta_{t+2}^V) + \cdots) \\
&= (1-\lambda)(\delta_t^V(1 + \lambda + \lambda^2 + \cdots) + \gamma\delta_{t+1}^V(\lambda + \lambda^2 + \lambda^3 + \cdots) + \cdots) \\
&= \sum_{l=0}^{\infty} (\gamma\lambda)^l \, \delta_{t+l}^V
\end{aligned}
$$

  - GAE $\approx$ TD(λ) advantage estimator

Policy gradient revisited

- π_θ : parameterized policy
- Trajectory $\tau = (s_0, a_0, s_1, \dots, s_T)$
- Return $R_t = \sum_{k=t}^{T} \gamma^{k-t} r_k$

- Objective function $J(\theta) = \mathbb{E}_{s_0 \sim p_0}[V^{\pi_\theta}(s_0)] \propto \mathbb{E}_{s \sim d^{\pi_\theta}}[V^{\pi_\theta}(s)]$
- Policy gradient: $\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim p_0, \pi_\theta}[\nabla_\theta \log P(\tau;\theta)\, R(\tau)]$
  $\propto \mathbb{E}_{s \sim d^{\pi_\theta}, a \sim \pi_\theta}[\nabla_\theta \log \pi_\theta(a|s)\, Q^{\pi}(s,a)]$

- REINFORCE (Monte-Carlo approach)

$$\nabla_\theta J(\theta) = \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t)\,(R_t - V(s_t)), \qquad R_t = \sum_{k=0}^{T-t-1} \gamma^k r_{t+k}$$

- Advantage Actor-Critic $\nabla_\theta J(\theta) \propto \nabla_\theta \log \pi_\theta(a|s)\,\delta,\ \delta = r + \gamma V_v(s') - V(s)$
  - stepwise update : $\min_\theta [-\log \pi_\theta(a|s)\,\delta]$
  - possible to update episode-wise: $\min_\theta \frac{1}{|\tau|} \sum_{(s,a,r,s') \in \tau} [-\log \pi_\theta(a|s)\,\delta]$

TRPO (Schulman et al, ICML 2015)

Trust Region Policy Optimization

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- Motivation
  - PG methods performance is not satisfactory
    - compared to derivative free methods such as CEM or CMA
    - Why? Step size matters!!
      - smaller step ➔ slow convergence
      - bigger step ➔ bad policy
        ➔ next episode under bad policy ➔ get worse
      - cf. supervised learning → label data corrects
  - Desiderata for policy optimization method
    - stable monotonic improvement of policy : how to choose step size?
    - good sample efficiency

Relative Policy Performance Identity

- Kakade and Langford, "Approximately Optimal Approximate RL", ICML 2002
- Relative policy performance identity
  - $J(\pi) = J(\pi_o) + \mathbb{E}_{\tau \sim p_0, \pi}\left[\sum_{t=0}^{\infty} \gamma^t A^{\pi_o}(s_t, a_t)\right]$

- **[Proof]**

$\tau := (s_0, a_0, s_1, a_1, \dots),\ A^{\pi_o}(s_t, a_t) = \mathbb{E}_{r_t, s_{t+1} \sim P(\cdot|s_t, a_t)}[r_t + \gamma V^{\pi_o}(s_{t+1})] - V^{\pi_o}(s_t)$

$\mathbb{E}_{\tau \sim p_0, \pi}[\sum_{t=0}^{\infty} \gamma^t A^{\pi_o}(s_t, a_t)]$

$= \mathbb{E}_{\tau \sim p_0, \pi}[\sum_{t=0}^{\infty} \gamma^t (r_t + \gamma V^{\pi_o}(s_{t+1}) - V^{\pi_o}(s_t))]$

$= \mathbb{E}_{\tau \sim p_0, \pi}[(\sum_{t=0}^{\infty} \gamma^t r_t) + (\gamma V^{\pi_o}(s_1) - V^{\pi_o}(s_0)) + (\gamma^2 V^{\pi_o}(s_2) - \gamma V^{\pi_o}(s_1)) + \cdots]$

$= \mathbb{E}_{\tau \sim p_0, \pi}[\sum_{t=0}^{\infty} (\gamma^t r_t) - V^{\pi_o}(s_0)]$

$= \mathbb{E}_{\tau \sim p_0, \pi}[\sum_{t=0}^{\infty} (\gamma^t r_t)] - \mathbb{E}_{s_0 \sim p_0(\cdot)}[V^{\pi_o}(s_0)]$

$= J(\pi) - J(\pi_o)$

Relative Policy Performance Identity

- Changing view point : sum over time steps → sum over states
  - $J(\pi) - J(\pi_o) = \mathbb{E}_{\tau \sim p_0, \pi}\left[\sum_{t=0}^{\infty} \gamma^t A^{\pi_o}(s_t, a_t)\right]$
  - $J(\pi) - J(\pi_o) = \sum_s \rho^{\pi}(s) \sum_a \pi(a|s) A^{\pi_o}(s,a) \propto \mathbb{E}_{s \sim d^{\pi}, a \sim \pi}\left[A^{\pi_o}(s,a)\right]$
  - ✓ $d^{\pi}(s)$: discounted stationary distribution
  - ✓ $\mathbb{E}_{\tau \sim p_0, \pi}\left[\sum_{t=0}^{\infty} \gamma^t A^{\pi_o}(s_t, a_t)\right]$

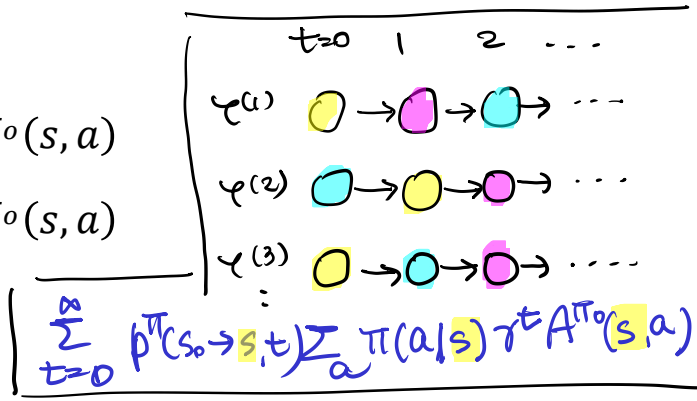
$$= \sum_{s_0} p_0(s_0) \sum_s \sum_{t=0}^{\infty} p^{\pi}(s_0 \to s, t) \sum_a \pi(a|s) \gamma^t A^{\pi_o}(s,a)$$

$$= \sum_{s_0} p_0(s_0) \sum_s \sum_{t=0}^{\infty} \gamma^t p^{\pi}(s_0 \to s, t) \sum_a \pi(a|s) A^{\pi_o}(s,a)$$

$$= \sum_{s_0} p_0(s_0) \sum_s \rho^{\pi}(s_0 \to s) \sum_a \pi(a|s) A^{\pi_o}(s,a)$$

$$= \sum_s \sum_{s_0} p_0(s_0) \rho^{\pi}(s_0 \to s) \sum_a \pi(a|s) A^{\pi_o}(s,a)$$

$$= \sum_s \rho^{\pi}(s) \sum_a \pi(a|s) A^{\pi_o}(s,a) \propto \sum_s d^{\pi}(s) \sum_a \pi(a|s) A^{\pi_o}(s,a) = \mathbb{E}_{s \sim d^{\pi}, a \sim \pi}\left[A^{\pi_o}(s,a)\right]$$



- ✓ Interpretation of $J(\pi) - J(\pi_o) \propto \mathbb{E}_{s \sim d^{\pi}, a \sim \pi}\left[A^{\pi_o}(s,a)\right]$
  - ✓ $\mathbb{E}_{s \sim d^{\pi}, a \sim \pi}\left[A^{\pi_o}(s,a)\right]$: advantage of π over π_o
  - We can improve performance $J(\pi)$ by increasing $\pi(a|s)$ where $A^{\pi_o}(s,a)$ is large

Surrogate objective

- relative performance objective
  - π_θ : policy we want to optimize. Let $\rho^{\pi_\theta} = \rho_\theta,\ d^{\pi_\theta} = d_\theta$
  - θ_{cur} : current policy parameter. Let $\pi_{\theta_{\text{cur}}} = \pi_{\text{cur}},\ \rho^{\pi_{\theta_{\text{cur}}}} = \rho_{\text{cur}},\ d^{\pi_{\theta_{\text{cur}}}} = d_{\text{cur}}$
  - $J(\pi_\theta) - J(\pi_{\text{cur}}) = \sum_s \rho_\theta(s) \sum_a \pi_\theta(a|s)\, A^{\pi_{\text{cur}}}(s,a) \propto \mathbb{E}_{s\sim d_\theta, a\sim\pi_\theta}\left[A^{\pi_{\text{cur}}}(s,a)\right]$
  - $\Delta(\pi_\theta;\pi_{\text{cur}}) = \mathbb{E}_{s\sim d_\theta, a\sim\pi_\theta}\left[A^{\pi_{\text{cur}}}(s,a)\right]$
    - maximizing $J(\pi_\theta) \equiv$ maximizing $\Delta(\pi_\theta;\pi_{\text{cur}})$
    - $\nabla_\theta \Delta(\pi_\theta;\pi_{\text{cur}}) = \mathbb{E}_{s\sim d_\theta, a\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a|s)\, A^{\pi_{\text{cur}}}(s,a)]$
    - **hard to optimize** because d_θ also depends on θ
- surrogate objective $L(\pi_\theta;\pi_{\text{cur}}) \cong \Delta(\pi_\theta;\pi_{\text{cur}})$

$L(\pi_\theta;\pi_{\text{cur}}) = \mathbb{E}_{s\sim d_{\text{cur}}, a\sim\pi_\theta}\left[A^{\pi_{\text{cur}}}(s,a)\right] = \sum_s d_{\text{cur}}(s) \sum_a \pi_\theta(a|s)\, A^{\pi_{\text{cur}}}(s,a)$

$\Rightarrow\ L(\pi_\theta;\pi_{\text{cur}})|_{\theta=\theta_{\text{cur}}} = \Delta(\pi_\theta;\pi_{\text{cur}})|_{\theta=\theta_{\text{cur}}}$

$\nabla_\theta L(\pi_\theta;\pi_{\text{cur}}) = \sum_s d_{\text{cur}}(s) \sum_a \nabla_\theta \pi_\theta(a|s)\, A^{\pi_{\text{cur}}}(s,a) = \mathbb{E}_{s\sim d_{\text{cur}}, a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s)\, A^{\pi_{\text{cur}}}(s,a)\right]$

$\nabla_\theta L(\pi_\theta;\pi_{\text{cur}})|_{\theta=\theta_{\text{cur}}} = \nabla_\theta \Delta(\pi_\theta;\pi_{\text{cur}})|_{\theta=\theta_{\text{cur}}}$

$L(\pi_\theta;\pi_{\text{cur}}) = \mathbb{E}_{s\sim d_{\text{cur}}, a\sim\pi_\theta}\left[A^{\pi_{\text{cur}}}(s,a)\right] \Rightarrow s\sim\rho_{\text{cur}}, a\sim\pi_\theta$: hard to sample from

$L(\pi_\theta;\pi_{\text{cur}}) = \mathbb{E}_{s\sim d_{\text{cur}}, a\sim\pi_{\text{cur}}}\left[\dfrac{\pi_\theta(s|s)}{\pi_{\text{cur}}(a|s)} A^{\pi_{\text{cur}}}(s,a)\right] \Leftarrow$ use importance sampling

$\Rightarrow$ we can use samples obtained by π_{cur}

Optimization within Trust Region

- Maximizing surrogate objective
  - maximizing $J(\pi_\theta) \equiv$ maximizing $\Delta(\pi_\theta; \pi_{\text{cur}})$
    $\cong$ maximizing $L(\theta) = L(\pi_\theta; \pi_{\text{cur}})$
  - this argument is valid only when θ is in the vicinity of θ_{cur}
    - how to define 'vicinity of θ_{cur}'?
    - trust region : region of θ where the policy π_θ is similar to π_{cur}
    - distance between two distributions : KL divergence, Wasserstein distance, …
      - $D_{KL}(p, q) = \sum_x p(x) \log \frac{p(x)}{q(x)}$
- TRPO - basic : Use constrained optimization (not Grad. Ascent)
  - $\max_\theta L(\pi_\theta; \pi_{\text{cur}})$
  - s.t. $\max_s D_{KL}\left(\pi_{\text{cur}}(.|s), \pi_\theta(.|s)\right) \le \delta$: Trust region (stay close to $\pi_{\text{cur}}(.|s)$)
- TRPO - sample based : change constraint into expectation form
  - $\max_\theta L(\theta) = \mathbb{E}_{s\sim d_{\text{cur}}, a\sim\pi_{\text{cur}}}\left[\frac{\pi_\theta(s|s)}{\pi_{\text{cur}}(a|s)} A^{\pi_{\text{cur}}}(s, a)\right]$: objective using samples
  - s.t. $\mathbb{E}_{s\sim d_{\text{cur}}}\left[D_{KL}\left(\pi_{\text{cur}}(.|s), \pi_\theta(.|s)\right)\right] \le \delta$: Trust region using sample mean (not max)

TRPO - sample based

$$\max_\theta L(\theta) = \mathbb{E}_{s\sim d_{\text{cur}}, a\sim\pi_{\text{cur}}}\left[\frac{\pi_\theta(a|s)}{\pi_{\text{cur}}(a|s)} A^{\pi_{\text{cur}}}(s,a)\right]$$
$$\text{s.t.}\ \ \mathbb{E}_{s\sim d_{\text{cur}}} D_{KL}\big(\pi_{\text{cur}}(.|s), \pi_\theta(.|s)\big) \le \delta$$

- Evaluating KL

$$\tau = (s_0, a_0, r_0, s_1, a_1, \dots, s_T)$$

$$p(\tau;\theta) = p_0(s)\prod_{t=0}^{T-1} p(s_{t+1}|s_t,a_t)\cdot\pi_\theta(a_t|s_t)$$

$$D_{KL}(P(\tau;\theta_{\text{cur}}), P(\tau;\theta)) = \sum_\tau P(\tau;\theta_{\text{cur}}) \log\frac{P(\tau;\theta_{\text{cur}})}{P(\tau;\theta)}$$

$$= \sum_\tau P(\tau;\theta_{\text{cur}}) \log\frac{p_0(s_0)\prod_{t=0}^{T-1} p(s_{t+1}|s_t,a_t)\cdot\pi_{\text{cur}}(a_t|s_t)}{p_0(s_0)\prod_{t=0}^{T-1} p(s_{t+1}|s_t,a_t)\cdot\pi_\theta(a_t|s_t)}$$

$$= \sum_\tau P(\tau;\theta_{\text{cur}}) \log\frac{\prod_{t=0}^{T-1}\pi_{\text{cur}}(a_t|s_t)}{\prod_{t=0}^{T-1}\pi_\theta(a_t|s_t)} = \sum_\tau P(\tau;\theta_{\text{cur}}) \sum_{t=0}^{T-1}\log\frac{\pi_{\text{cur}}(a_t|s_t)}{\pi_\theta(a_t|s_t)}$$

$$\cong \frac{1}{N}\sum_{a_t,s_t\sim\pi_{\text{cur}}} \log\frac{\pi_{\text{cur}}(a_t|s_t)}{\pi_\theta(a_t|s_t)}$$

$$\max_\theta L(\theta) = \frac{1}{N}\sum_\tau\sum_{a_t,s_t\in\tau}\left[\frac{\pi_\theta(a_t|s_t)}{\pi_{\text{cur}}(a_t|s_t)} A^{\pi_{\text{cur}}}(a_t|s_t)\right]$$
$$\text{s.t.}\quad \frac{1}{N}\sum_\tau\sum_{a_t,s_t\in\tau}\log\frac{\pi_{\text{cur}}(a_t|s_t)}{\pi_\theta(a_t|s_t)} \le \delta$$

Practical TRPO algorithm

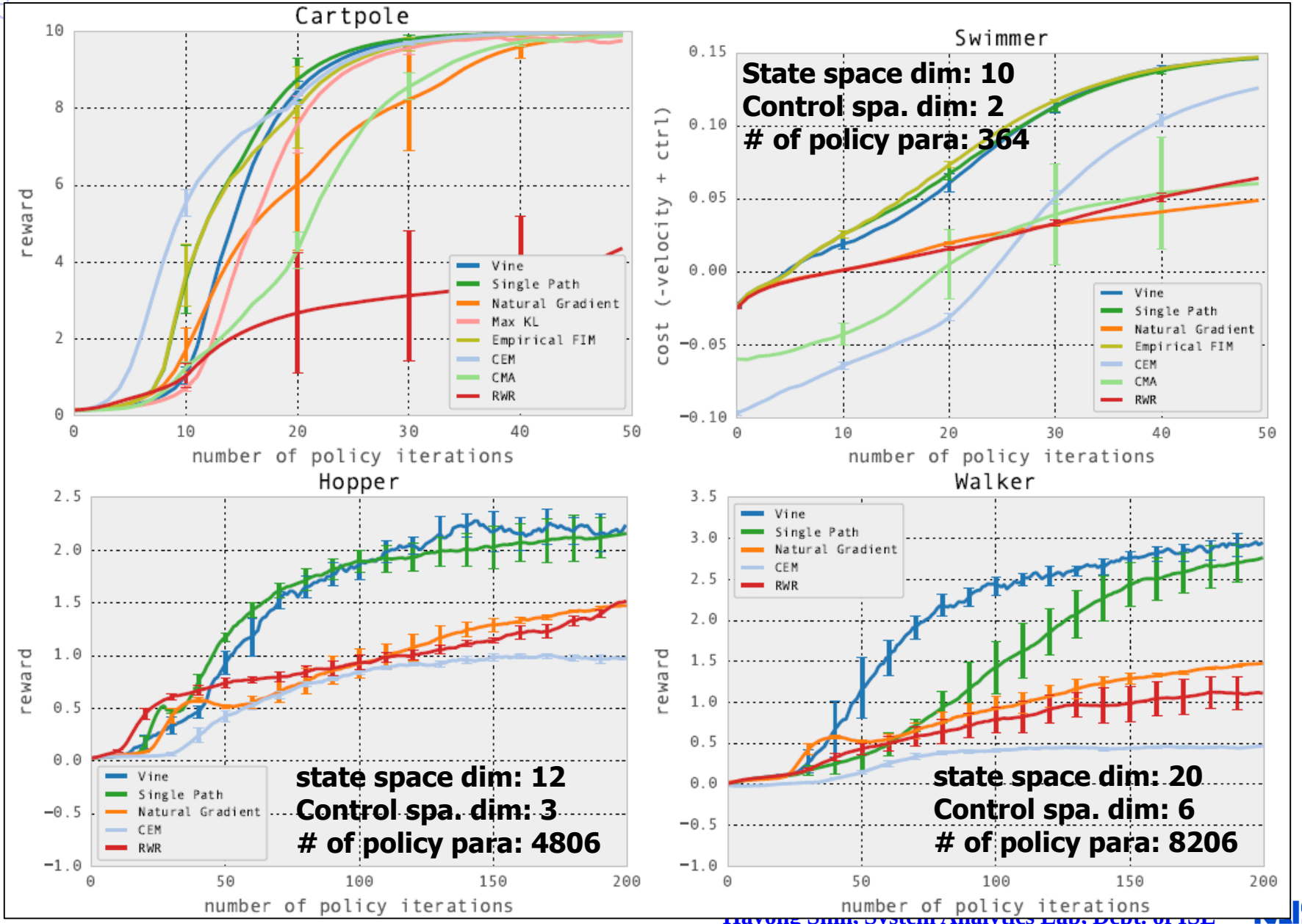
➢ **Algorithm TRPO**

1. initialize policy π_θ $(\theta_{\text{cur}} = \theta_0)$ and state value V_v $(v = v_0)$
2. for iteration = 0,1,…
   - sample trajectories $D = \{\tau_i; i = 1..N\}$ from $\pi_{\theta_{\text{cur}}}$
   - for each τ_i, compute return R_t and advantage A_t at each time step t
     - $R_t = \sum_{k=t}^{T} \gamma^{k-t} r_k, \quad A_t = R_t - V_v(s_t)$
   - compute policy gradient for D_k
     - $g_k = \frac{1}{N} \sum_\tau \sum_t \frac{\nabla_\theta \pi_\theta(a|s)}{\pi_{\theta_{\text{cur}}}(a|s)} A_t = \frac{1}{N} \sum_\tau \sum_t \nabla_\theta \log \pi_\theta(a|s) \,|_{\theta=\theta_{\text{cur}}} \; A_t$
   - update θ_{cur} within Trust Region δ
     - compute Hessian of sample average KL divergence of $\pi_\theta(a|s)$
     - conjugate gradient and backtracking line search
   - refit value function $V_v(s)$ to D
     - by minimizing $\frac{1}{N} \sum_\tau \sum_t \left(R_t - V_v(s_t)\right)^2$

➢ Computing A_t can take other form, incl. TD error $A_t = r_t + \gamma V_v(s_{t+1}) - V_v(s_t)$

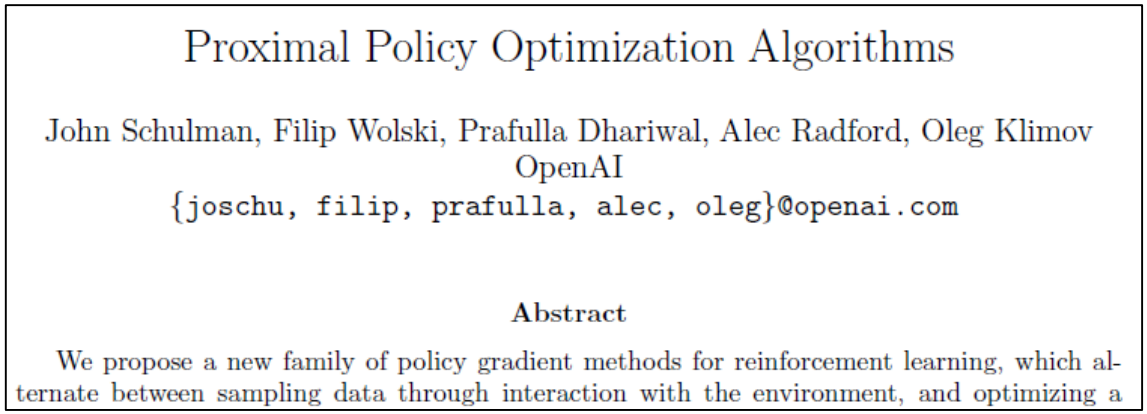
➢ Computing Hessian → expensive and unstable in many cases

Experiment Results1_Robotic Locomotion



PPO (Schulman et al, arXiv 2017)

- Schulman et al. "Proximal policy optimization algorithms." arXiv 2017



Proximal Policy Optimization Algorithms

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Abstract

We propose a new family of policy gradient methods for reinforcement learning, which alternate between sampling data through interaction with the environment, and optimizing a

- Motivation
  - TRPO utilizes Hessian (2nd order derivative)
    - unstable and expensive
  - can we replace it with 1st order approximation?

- PPO
  - Easy to implement
  - Fast (no Hessian required)
  - Good performance

"Proximal" Policy Optimization

- TRPO

$$\max_{\theta} \frac{1}{N}\sum_{n=1}^{N}\left[\frac{\pi_\theta(a_n|s_n)}{\pi_{\text{cur}}(a_n|s_n)}A_n\right] \quad s.t. \quad \overline{D_{KL}}(\pi_{\text{cur}},\pi_\theta) \le \delta$$

$$\overline{D_{KL}}(\pi_{\text{cur}},\pi_\theta) = \frac{1}{N}\sum_{a_n,s_n\sim\pi_{\theta_{\text{cur}}}} \log\frac{\pi_{\text{cur}}(a_n|s_n)}{\pi_\theta(a_n|s_n)}$$

: sample average of KL divergence

$\overline{D_{KL}}(\pi_{\text{cur}},\pi_\theta) \le \delta$: constrains parameter change close to θ_{cur}

- PPO-Penalty : Lagrange relaxation of trust region constraint
  - Perform SGD on the following (KL penalized) objective function

$$\max_{\theta} \frac{1}{N}\sum_{n=1}^{N}\left[\frac{\pi_\theta(a_n|s_n)}{\pi_{\text{cur}}(a_n|s_n)}A_n\right] - \beta\,\overline{D_{KL}}(\pi_{\text{cur}},\pi_\theta)$$

  - adaptively adjust β depending on $d = \overline{D_{KL}}(\pi_{\text{cur}},\pi_\theta)$
    - if $d < 0.67 \cdot d_{target}$, $\beta \leftarrow \beta/2$ (too small change → decrease penalty weight)
    - if $d > 1.5 \cdot d_{target}$, $\beta \leftarrow \beta * 2$ (too big change → increase penalty weight)

"Proximal" Policy Optimization

- PPO-Clip (Clipped PPO)
  - Use "Clipping" on objective : instead of using KL divergence

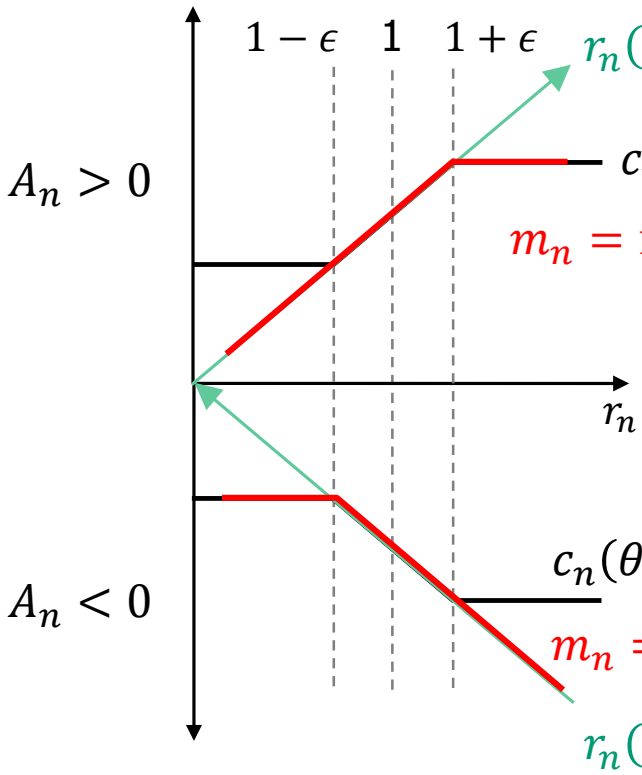
$$L(\theta) = \frac{1}{N}\sum_{n=1}^{N}[r_n(\theta)A_n]\,, \text{where } r_n(\theta) = \frac{\pi_\theta(a_n|s_n)}{\pi_{\text{cur}}(a_n|s_n)} \in [0,\infty]$$

$c_n(\theta) = clip(r_n(\theta), 1-\epsilon, 1+\epsilon) \in [1-\epsilon, 1+\epsilon]$: clipped weight

$$m_n(\theta) = min\{r_n(\theta)A_n, c_n(\theta)A_n\}$$

$$L^{clip}(\theta) = \frac{1}{N}\sum_{n=1}^{N} m_n(\theta)$$

$$m_n(\theta) = min\{r_n(\theta)A_n, c_n(\theta)A_n\} = \begin{cases} (1+\epsilon)A_n, \text{if } A_n > 0 \text{ and } r_n(\theta) > 1+\epsilon \\ r_n(\theta)A_n, \text{if } A_n \geq 0 \text{ and } r_n(\theta) \leq 1+\epsilon \\ r_n(\theta)A_n, \text{if } A_n \leq 0 \text{ and } r_n(\theta) \geq 1-\epsilon \\ (1-\epsilon)A_n, \text{if } A_n < 0 \text{ and } r_n(\theta) < 1-\epsilon \end{cases}$$



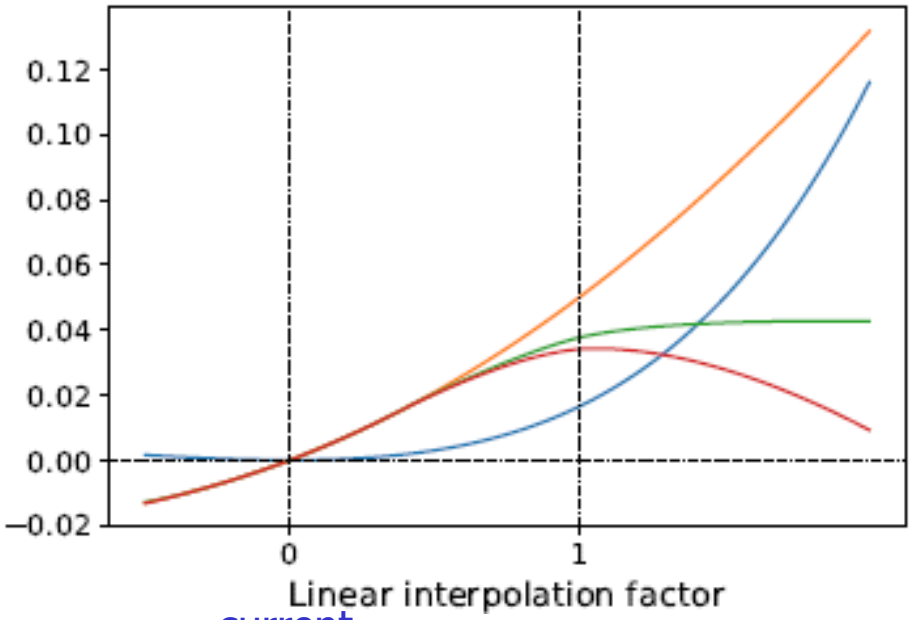
- if $A_n > 0$, increase $\pi_\theta(a_n|s_n)$
- $m_n(\theta)$ is limited to $(1+\epsilon)A_n$

➔ no motivation to increase beyond this

- if $A_n < 0$, decrease $\pi_\theta(a_n|s_n)$
- $m_n(\theta)$ is limited to $(1-\epsilon)A_n$

➔ no motivation to decrease beyond this

- Consider $N = 2, A_1 = -1, A_2 = +2$
  - Assume : $\theta \uparrow$ ➔ $r_1(\theta) \uparrow$ and $r_2(\theta) \uparrow$ (at similar rates)
  - $L^{CPI} = (r_1(\theta)A_1 + r_2(\theta)A_2)/2$ increases as θ increases
  - both $c_1(\theta)A_1$and $c_2(\theta)A_2$ get flat when θ goes beyond some limit
  - $m_1(\theta) = \min\{c_1(\theta)A_1, r_1(\theta)A_1\}$: gets flat
  - $m_2(\theta) = \min\{c_2(\theta)A_2, r_2(\theta)A_2\}$: keeps decreasing
  - $L^{CLIP} = (m_1(\theta) + m_2(\theta))/2$ attains maximum around the limit



current policy

limit

$\hat{E}_t[KL_t]$ mean KL div.

$L^{CPI} = \hat{E}_t[r_t A_t]$ objective in TRPO

$\hat{E}_t[\text{clip}(r_t, 1-\varepsilon, 1+\varepsilon)A_t]$ imp. weight clipping

$L^{CLIP} = \hat{E}_t[\min(r_t A_t, \text{clip}(r_t, 1-\varepsilon, 1+\varepsilon)A_t)]$

taking min.

TAQ. So can we use large step size?

TAQ. Can use backtracking line search?

PPO-Clip Algorithm

➢ **Algorithm PPO-Clip (Basic)**

1. initialize policy π_θ $(\theta = \theta_0)$ and state value V_v $(v = v_0)$
2. for iteration = 0,1,…
   - $\theta_{\text{cur}} \leftarrow \theta$
   - generate an episode τ from $\pi_{\theta_{\text{cur}}}$ (up to T time steps for continuing task)
     - at each time step, store $\text{seg}_t = (s_t, a_t, r_t, s_{t+1}, \log \pi_{\text{cur}}(a_t|s_t))$ to τ
     - compute return R_t and advantage A_t at each time step t
       - $R_t = \sum_{k=t}^{T-1} \gamma^{k-t} r_k$
       - $A_t = R_t - V_v(s_t)$ (or TD adv. $\delta_t^V = r_t + \gamma V_v(s_{t+1}) - V_v(s_t)$, GAE $\sum_{l=0}^{T-t-1} (\gamma\lambda)^l \, \delta_{t+l}^V$)
       - append R_t and A_t to seg_t
   - update parameters
     - update θ by maximizing $L^{clip}(\theta)$: $\theta \leftarrow \theta + \alpha \nabla L^{clip}(\theta)$
       - $L^{clip}(\theta) = \frac{1}{|\tau|} \sum_{\text{seg}_n \in \tau} min\{r_n(\theta) A_n, \; c_n(\theta) A_n\}$
         - » $r_n(\theta) = \frac{\pi_\theta(a_n|s_n)}{\pi_{\text{cur}}(a_n|s_n)} = exp(\log \pi_\theta(a_n|s_n) - \log \pi_{\text{cur}}(a_n|s_n))$
         - » $c_n(\theta) = clip(r_n(\theta), 1-\epsilon, 1+\epsilon)$
       - update v by minimizing $L^{VF}(v) = \frac{1}{|\tau|} \sum_{\text{seg}_n \in \tau} (V_v(s_n) - R_t)^2$

PPO-Clip Algorithm

TAQ. Is shared net better?

➢ Algorithm PPO-Clip (adv.)

1. initialize policy π_θ and state value V_θ $(\theta = \theta_0)$: V_θ and π_θ share parameter (2 heads net)
2. for k= 0,1,…
   - $\theta_{\text{cur}} \leftarrow \theta$
   - for N episodes : generate episodes
     - generate an episode τ from π_{cur} (or up to T time steps)
       - at each time step, store $\text{seg}_t = (s_t, a_t, r_t, s_{t+1}, \log \pi_{\text{cur}}(a_t|s_t))$ to RB
     - for episode τ, compute return R_t and advantage A_t at each time step t
       - $R_t = \sum_{k=t}^{T-1} \gamma^{k-t} r_k$, $A_t = R_t - V_v(s_t)$ or $A_t = \sum_{l=0}^{T} (\gamma\lambda)^l \delta_{t+l}^V$ (GAE)
       - append R_t and A_t to seg_t
   - for E epochs : update parameters
     - sample a minibatch MB ⊂ RB
     - update θ by minimizing $L(\theta) = -L^{clip}(\theta) + c_1 L^{VF}(\theta) - c_2 L^{ent}(\theta)$
       - $L^{clip}(\theta) = L^{clip}(\theta) = \frac{1}{|MB|} \sum_{\text{seg}_n \in MB} min\{r_n(\theta)A_n,\ c_n(\theta)A_n\}$
         - » $r_n(\theta) = \frac{\pi_\theta(a_n|s_n)}{\pi_{\text{cur}}(a_n|s_n)} = exp(\log \pi_\theta(a_n|s_n) - \log \pi_{\text{cur}}(a_n|s_n))$
         - » $c_n(\theta) = clip(r_n(\theta), 1-\epsilon, 1+\epsilon)$
       - $L^{VF}(\theta) = \frac{1}{|MB|} \sum_{\text{seg}_n \in MB} (V_\theta(s_n) - R_t)^2$
       - $L^{ent}(\theta) = -\sum_{\text{seg}_n \in MB} \sum_a \pi_\theta(a|s_n) \log \pi_\theta(a|s_n)$: entropy of $\pi_\theta(a|s_n)$ for exploration
   - clear RB

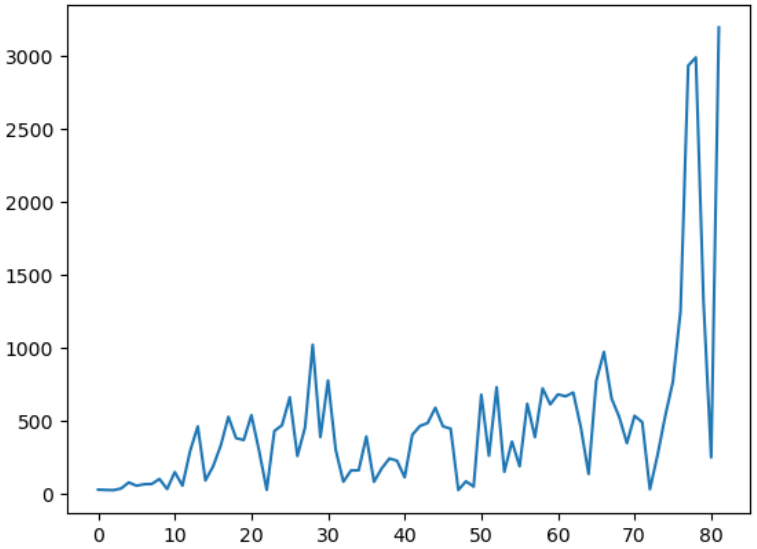
TAQ. Is PPO on-policy or off-policy?

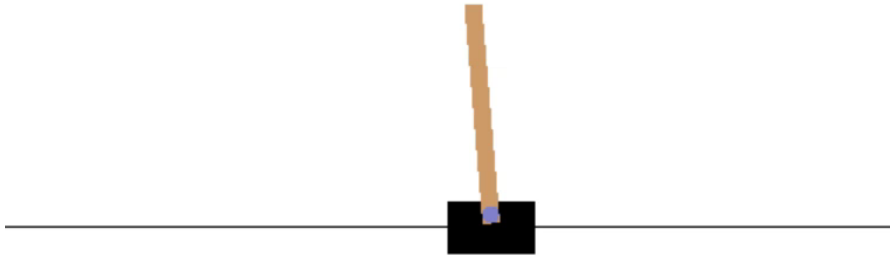
- off-policy?
  - PPO looks like off-policy since experience is generated using π_{cur} while optimizing π_θ



PPO for Cartpole

- Cartpole-v1 with angle reward
- PPO
  - can be combined with GAE : a little better performance
  - Reached max steps (2000) after training 85 episodes





Experiment: Continuous Domain

- **MuJoCo 7 games**



Figure 3: Comparison of several algorithms on several MuJoCo environments, training for one million timesteps.